Journal of Electromyography and Kinesiology 23 (2013) 640–648

Contents lists available at SciVerse ScienceDirect

Journal of Electromyography and Kinesiology

journal homepage: www.elsevier.com/locate/jelekin

Age-related changes in neuromuscular function of the quadriceps muscle in physically active adults

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ARTICLE INFO

Article history:
Received 5 July 2012
Received in revised form 10 December 2012
Accepted 10 January 2013

Keywords:
Aging
Quadriceps muscle
Dynapenia
Voluntary activation
Interpolated twitch technique
Electromyography
Spinal excitability
H reflex
M wave
Contractile properties
Twitch contraction

ABSTRACT

Substantial evidence exists for the age-related decline in maximal strength and strength development. Despite the importance of knee extensor strength for physical function and mobility in the elderly, studies focusing on the underlying neuromuscular mechanisms of the quadriceps muscle weakness are limited.

The aim of this study was to investigate the contributions of age-related neural and muscular changes in the quadriceps muscle to decreases in isometric maximal voluntary torque (iMVT) and explosive voluntary strength. The interpolated twitch technique and normalized surface electromyography (EMG) signal during iMVT were analyzed to assess changes in neural drive to the muscles of 15 young and 15 elderly volunteers. The maximal rate of torque development as well as rate of torque development, impulse and neuromuscular activation in the early phase of contraction were determined. Spinal excitability was estimated using the *H* reflex technique. Changes at the muscle level were evaluated by analyzing the contractile properties and lean mass.

The age-related decrease in iMVT was accompanied by a decline in voluntary activation and normalized surface EMG amplitude. Mechanical parameters of explosive voluntary strength were reduced while the corresponding muscle activation remained primarily unchanged. The spinal excitability of the vastus medialis was not different while *M* wave latency was longer. Contractile properties and lean mass were reduced.

In conclusion, the age-related decline in iMVT of the quadriceps muscle might be due to a reduced neural drive and changes in skeletal muscle properties. The decrease in explosive voluntary strength seemed to be more affected by muscular than by neural changes.

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1. Introduction

The age-related decrease in muscle strength (dynapenia) and strength development is well-documented and might be caused by neural modulations and alterations of skeletal muscle properties (Clark and Taylor, 2011; Manini and Clark, 2011). However, the results differ in part between the investigated muscle groups possibly due to differences in the function and physiological profile of the muscles (Clark and Taylor, 2011).

Numerous studies have described the modulations of voluntary strength and morphology of the quadriceps muscle with aging (Porter et al., 1995; Roos et al., 1999). Due to its functional importance for gait (Murdock and Hubley-Kozey, 2012) and maintaining posture (Barbeau et al., 2000), it is surprising that studies focusing on the underlying neuromuscular mechanisms of the knee extensor weakness are limited. Moreover, the reported results are partially conflicting, particularly with regard to changes in neural drive to the quadriceps muscle (Harridge et al., 1999; Roos et al., 1999; Stevens et al., 2003; Wilder and Cannon, 2009). The ability to voluntarily activate the knee extensors during isometric contractions has been estimated using either the interpolated twitch technique or the central activation ratio. The inconsistent findings may be primarily related to methodological limitations and differences of these two techniques (Klass et al., 2007). Thus, the simultaneous application of two different methods may provide greater insight into modulations of neural drive to the muscle. Only one study has considered a twofold approach to detecting age-related changes in efferent motoneuron output to the knee extensors (Cannon et al., 2007). Besides the interpolated twitch technique, Cannon et al. have used normalized muscle activity during isometric maximal strength to estimate the neural drive. Both parameters were not changed indicating no differences in efferent motoneuron output to the quadriceps muscle. However, the authors used the standard formula for estimating voluntary activation. It has been demonstrated that the interpolated twitch technique has its limitations when the standard formula is applied (Folland and Williams, 2007). The moment when the stimulus is delivered is not always the time point of maximal strength. Thus, the use of a corrected

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http://dx.doi.org/10.1016/j.jelekin.2013.01.009

formula which includes the variations of the torque level has been recommended (Strojnik and Komi, 1998). Only Suetta et al. (2009) applied this corrected formula and also demonstrated no age-related change in voluntary activation under isometric conditions. However, these authors did not use a second method to assess the neural drive to the muscle in order to confirm their results.

It is further noteworthy that age-related changes in neural drive to the quadriceps during strength development have been less frequently investigated. Dynapenia of the quadriceps is a determinant of fall risk (Lord et al., 1994) and mortality (Newman et al., 2006). Considering the fact that the capacity to activate the muscle quickly may be more essential for maintaining balance and preventing falls than the maximal strength (Schultz, 1995), neuromuscular modulations of the quadriceps muscle during strength development should be investigated more closely.

In summary, there is some lack of knowledge about the neuromuscular mechanisms behind the modulations of maximal strength and strength development of the quadriceps muscle with advancing age. The aim of the present study was to investigate the contributions of age-related neural and muscular changes to decreases in isometric maximal voluntary torque (iMVT) and explosive voluntary strength. Explosive voluntary strength was analyzed by calculating the maximal rate of torque development (MRTD) as well as the rate of torque development (RTD) and impulse (IMP) in the early phase of contraction in time intervals of 0–50 ms, 50–100 ms, 100–150 ms and 150–200 ms after torque onset. The neural drive to the muscle during iMVT was estimated using two approaches: the interpolated twitch technique (corrected formula) and the root mean square of the EMG signal (RMS–EMG) normalized to maximal *M* wave (M_{max}). The neural drive to the muscle during MRTD and RTD was assessed using the RMS–EMG normalized to the M_{max} and normalized to the RMS–EMG during iMVT. Age-related changes in α-motoneuron excitability via Ia afferents were determined with the *H* reflex technique. Modulations at the skeletal muscle level were evaluated by analyzing the twitch torque signal induced by supramaximal electrical stimulation and by determining the lean mass of the leg.

It was hypothesized that iMVT, MRTD, RTD and IMP were reduced with aging accompanied by changes in neural drive to the quadriceps muscle. It was further assumed that aging would affect spinal excitability, contractile properties and skeletal muscle mass.

2. Methods

2.1. Subjects

Thirty physically active and healthy subjects with no history of neurological and musculoskeletal disorders or injuries volunteered to participate in this study. Physically active and healthy subjects were chosen in order to take account of the problem of lack of motivation and disuse atrophy. The subjects were assigned to two groups: (1) young, (2) elderly, each with 15 participants who refrained from consuming caffeine or alcohol and performing strenuous leg exercise 24 h and 48 h preceding the experiment. Prior to participation written informed consent was obtained from all subjects. The study was conducted according to the declaration of Helsinki and approved by the local ethics committee (A 2009 52). Subject characteristics are displayed in Table 1.

2.2. Experimental procedure

The subjects participated in three experimental sessions. The first session included the measurement of body composition using dual-energy X-ray absorptiometry (DXA) followed by two sessions of neuromuscular tests: (1) familiarization, (2) experiment, separated by one week. Subjects were seated in a standardized position on a CYBEX NORM dynamometer. Prior to the experiment subjects sat passively for 15 min without any warm-up in order to avoid potentiation effects (Folland et al., 2008). The neuromuscular tests were carried out on the quadriceps muscle of the dominant leg. A schematic illustration of the neuromuscular tests is displayed in Fig. 1.

Table 1
Subject characteristics and lean mass of the leg. *Denotes a significant difference between the groups (*<0.05, **<0.01).

	Young (n = 15)	Elderly (n = 15)	p
Men, n (%)	8 (53.3)	8 (53.3)	
Age, yrs, Mean (SD)	25.3 (3.6)	69.6 (3.1)	0.000**
Weight, kg, Mean (SD)	71.3 (11.6)	72.6 (9.9)	0.750
Height, m, Mean (SD)	1.75 (0.9)	1.69 (0.9)	0.058
Physical activity, h/week, Mean (SD)	4.6 (3.6)	3.7 (3.3)	0.134
Lean mass, kg, Mean (SD)	9.27 (1.65)	7.79 (1.72)	0.028*

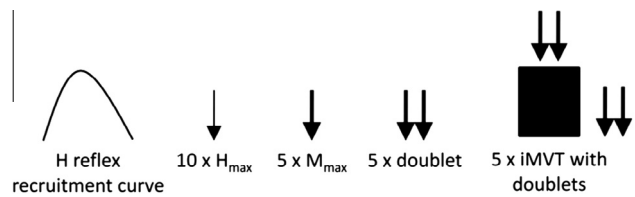

Fig. 1. Schematic illustration of neuromuscular tests. The *thin arrow* indicates stimulation at H_{max} intensity, the *thick arrow* indicates stimulation at supramaximal intensity, *double thick arrow* indicates doublet at supramaximal intensity.

2.3. Torque and EMG recordings

Torque signals were measured using a CYBEX NORM dynamometer (Computer Sports Medicine®, Inc., Stoughton, MA). The tests were performed in a sitting position with a knee angle of 75° and a hip angle of 50° (0° = full extension). The iMVT was tested by asking the subjects to exert isometric knee extensions as forcefully and as fast as possible against the lever arm of the dynamometer for 3 s. The maximal attempts were recorded until the coefficient of variance of five subsequent trials was below 5%. A rest period of 2 min was allowed between the trials.

The surface EMG was recorded using bipolar EMG Ambu® Blue Sensor N electrodes (2 cm diameter). The electrodes were applied to the shaved, abraded and cleaned skin over the middle of the muscle bellies of the vastus medialis, vastus lateralis and rectus femoris. Signals were amplified (2500×), band-pass filtered (10–450 Hz) and digitized with a sampling frequency of 5 kHz through an analog-to-digital converter (DAQ Card™-6024E, National Instruments, USA). The EMG and torque signals were analyzed using a custom built LABVIEW® based program (Imago, Pfisoft, Germany). A detailed description of EMG and torque measurements has been given previously (Behrens et al., 2012).

2.4. Transcutaneuous electrical stimulation

Transcutaneuous electrical stimulation was used to assess voluntary activation, spinal excitability and contractile properties of the quadriceps muscle. The anode (self-adhesive electrode, 35 × 45 mm, Spes Medica, Italy) was placed over the greater trochanter. The cathode (ball electrode, 1 cm diameter) was fixed to the subject's femoral triangle, 3–5 cm below the inguinal ligament.

One millisecond rectangular pulses (400 V) were applied to the femoral nerve by a constant-current stimulator (Digitimer® DS7A, Hertfordshire, UK). The inter stimulus intervals (ISI 10 s, i.e. 0.1 Hz) were provided by a Digitimer® train/delay generator (DG2A, Hertfordshire, UK). Different current intensities were randomly delivered to the femoral nerve to evoke the maximal H reflex (H_{max}) of the vastus medialis and M_{max} of the vastus medialis, rectus femoris and vastus lateralis. The H_{max} and M_{max} were elicited and recorded 5 and 10 times, respectively. The evaluation of contractile properties and voluntary activation was realized at a stimulation intensity of 40% above the level that was needed for eliciting a maximal twitch response and concomitant M_{max}. Resting twitch responses were evoked using single and doublet (ISI 10 ms, i.e. 100 Hz) stimulation. Voluntary activation was assessed using the interpolated twitch technique (Allen et al., 1995). A transcutaneous supramaximal doublet was given 2 s after torque onset, during the plateau phase. The second doublet was applied 2 s after the contraction.

2.5. Lean body mass

Leg lean body mass was assessed using DXA. A Lunar Prodigy densitometer (General Electric (GE) Medical System Lunar, Madison, WI, USA) measured the attenuation of X-rays (76 kV, 0.15 mA, 0.4 μGy) while a total body scan was performed in the supine position with arms at the side. The scan time took 6–7 min, using the standard speed mode (irradiation time 274 s). The quality assurance procedure was performed before each test using a cuboid calibration phantom (200 × 130 × 60 mm). DXA provides a measure of appendicular lean soft tissue (ALST) which includes skeletal muscle mass and other components of bone free lean mass (i.e. ligaments, tendons, joint capsula and meniscal tissue) (Kim et al., 2002). A large proportion of ALST is skeletal muscle mass. Levine et al. (2000) found a high correlation between DXA thigh lean mass and CT-determined skeletal muscle mass ($r = 0.86$, $p < 0.001$). Thus, ALST can be considered as an estimate of skeletal muscle mass.

2.6. Data analysis

2.6.1. IMVT, explosive voluntary strength and voluntary activation

The highest maximal voluntary contraction was retained for analysis. The following parameters were calculated: iMVT and MRTD, as well as RTD and IMP in the early contraction phase. MRTD was defined as the maximal slope of the torque-time curve. The early rise in muscle strength was obtained from the average slope of the torque-time curve in different time intervals after torque onset: 0–50 ms (RTD_{50}), 50–100 ms (RTD_{100}), 100–150 ms (RTD_{150}), 150–200 ms (RTD_{200}). Onset of contraction was determined when the torque increased 3.5 Nm above the resting baseline level (Suetta et al., 2007). RTD was normalized to iMVT (RTD/iMVT). Absolute and normalized RTD values were considered for analysis. IMP was defined as the area under the torque-time curve.

The RMS–EMG for the vastus medialis, vastus lateralis and rectus femoris was calculated during a 200 ms epoch at iMVT ($RMS–EMG_{iMVT}$), this means prior to the electrical stimuli when the maximal torque level was achieved at the moment when the superimposed stimuli were delivered or 100 ms preceding and 100 ms following the iMVT if the maximal torque level was reached before the superimposed stimuli were administered. The RMS–EMG was also calculated at MRTD ($RMS–EMG_{MRTD}$), (i.e., 50 ms on either side of MRTD). The $RMS–EMG_{iMVT}$ and $RMS–EMG_{MRTD}$ of each muscle were normalized to their respective M_{max} value and averaged across the vastus medialis, vastus lateralis and rectus femoris to provide a global index of total quadriceps activity ($\overline{X}RMS–EMG_{iMVT}/M_{max}$, $\overline{X}RMS–EMG_{MRTD}/M_{max}$).

Furthermore, the RMS–EMG was calculated in the early phase of contraction in time windows of 0–50, 50–100, 100–150 and 150–200 ms relative to the onset of the EMG signal ($RMS–EMG_{RTD}$). The $RMS–EMG_{RTD}$ values were normalized to M_{max} ($RMS–EMG_{RTD}/M_{max}$) as well as to the RMS–EMG during iMVT ($RMS–EMG_{RTD}/RMS–EMG_{iMVT}$). Absolute and normalized $RMS–EMG_{RTD}$ values were averaged across the three muscles ($\overline{X}RMS–EMG_{RTD}$, $\overline{X}RMS–EMG_{RTD}/M_{max}$, $\overline{X}RMS–EMG_{RTD}/RMS–EMG_{iMVT}$).

Voluntary activation was calculated using the formula: %VA = (1 − superimposed twitch × (Tb/iMVT)/control twitch) × 100, whereby Tb represents the torque level just before the superimposed twitch (Strojnik and Komi, 1998).

2.6.2. Spinal excitability

The amplitudes of H_{max} and M_{max} were measured peak-to-peak. H_{max} was normalized to M_{max} (H_{max}/M_{max}-ratio) in order to assess alterations in α-motoneuron excitability and/or presynaptic inhibition of primary muscle spindle afferents (Zehr, 2002). Furthermore, the latencies of H and M waves were calculated by determining the time interval between the stimulus artefact and the first deflection of the amplitudes. The latencies of H and M waves were measured to detect modulations in the signal conduction speed through the reflex arc and the direct motor pathway. The latencies were normalised to the height of the subjects (Δt_H/height, Δt_M/height) since the length of the reflex pathway is related to the subjects' height (Guiheneuc and Bathien, 1976).

2.6.3. Contractile properties and lean mass

The resting twitch torques were analyzed with regard (1) peak torque (PT), the highest value of the twitch torque signal, (2) maximal rate of torque development ($MRTD_{TT}$), the highest value of the first derivative of the twitch toque signal, (3) maximal rate of torque relaxation ($MRTR_{TT}$), the lowest value of the first derivative of the twitch toque signal, (4) twitch contraction time (TCT_{TT}), the time from the onset to the maximal twitch torque, (5) twitch half-relaxation time ($THRT_{TT}$), the time from the maximal twitch torque to one-half of its peak value and (6) total twitch area (TTA_{TT}), the area under the torque-time curve.

Lean mass of the dominant leg was calculated by the software Lunar *enCORE*™ 2007 (version 11.40.004). It has been shown that the reliability of regional ALST is high when machine-made regions of interest are manual adjusted (Lohman et al., 2009). In order to avoid measurement errors, the calculated regions were checked manually to make corrective adjustments if necessary.

2.7. Statistical analysis

Data were checked for normal distribution using a Shapiro–Wilk W test. Age-related differences between the groups were tested for significance by the unpaired Student's t test or Mann–Whitney U test. All data were analysed using the SPSS statistical package 19.0 (SPSS Inc., Chicago, IL, USA). The level of significance was established at $p < 0.05$. Data are presented as mean ± standard deviation in the tables and displayed as boxplots in the figures.

3. Results

3.1. IMVT, explosive voluntary strength and voluntary activation

IMVT and voluntary activation were decreased by 37.9% ($p < 0.001$) and 6.1% ($p = 0.034$), respectively, in the elderly subjects. $\overline{X}RMS–EMG_{iMVT}/M_{max}$ was significantly reduced by 16.4% ($p = 0.036$) (Fig. 2A–C).

Absolute MRTD and MRTD/iMVT were significantly lower (66.1%, $p < 0.001$ and 48.0%, $p < 0.001$) in the elderly compared with young subjects (Fig. 2D and E). No differences were found for normalized muscle activity during MRTD (Fig. 2F).

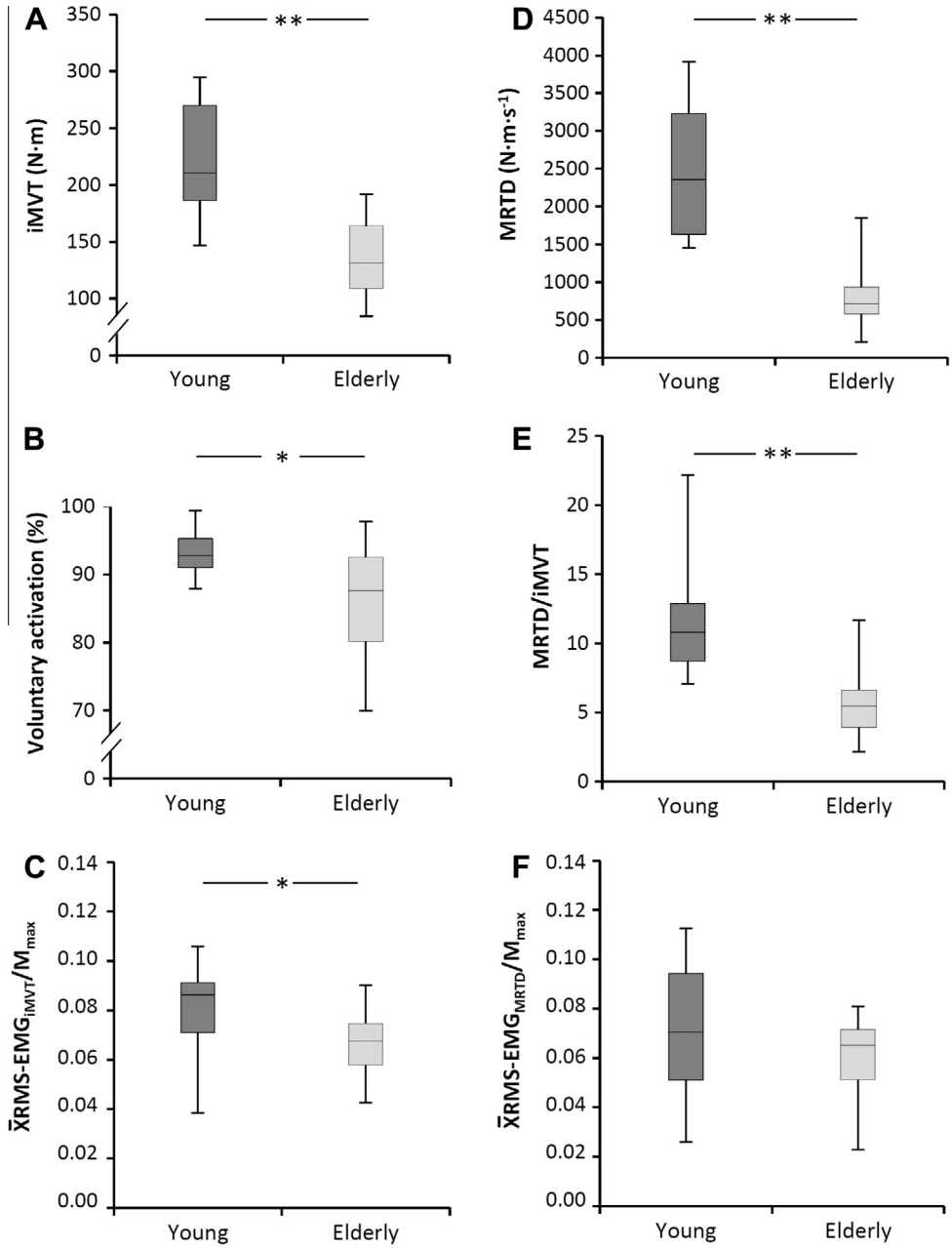

Fig. 2. Comparisons between groups. (A) isometric maximal voluntary torque (iMVT), (B) voluntary activation, (C) normalized and averaged root mean square of the EMG signal of vastus medialis, rectus femoris and vastus lateralis during iMVT ($\overline{X}$RMS–EMG$_{iMVT}$/M_{max}), (D) maximal rate of torque development (MRTD), (E) MRTD normalized to iMVT (MRTD/iMVT) and (F) normalized and averaged root mean square of the EMG signal of the three muscles during MRTD ($\overline{X}$RMS–EMG$_{MRTD}$/M_{max}). *Denotes a significant difference between the groups (* <0.05, **<0.01).

Absolute RTD and IMP were significantly reduced during all of the four 50 ms time windows (Fig. 3A and C). The highest reduction of absolute RTD (67.6%, $p = 0.003$) and IMP (73.2%, $p = 0.007$) was observed for the first 50 ms after torque onset. Afterwards, the difference in absolute RTD (45.8% RTD$_{100}$, $p = 0.002$, 34.4% RTD$_{150}$, $p = 0.015$, 35.8% RTD$_{200}$, $p = 0.001$) and IMP (60.6% IMP$_{100}$, $p < 0.001$, 53.8% IMP$_{150}$, $p < 0.001$, 52.6% IMP$_{200}$ $p < 0.001$) between the two groups decreased. By contrast, a decline in RTD/iMVT (48.18%, $p = 0.015$) was only observed in the first time window (0–50 ms) after torque onset (Fig. 3B).

$\overline{X}$RMS–EMG$_{RTD}$ was significantly lower during the 50–100 ms (44.1%, $p = 0.030$), 100–150 ms (43.0%, $p = 0.027$) and 150–200 ms (43.3%, $p = 0.010$) of the contraction phase in the elderly (Fig. 4A). On the contrary, there was a significant age-related increase of $\overline{X}$RMS–EMG$_{RTD}$/RMS–EMG$_{iMVT}$ by 27.0% ($p = 0.015$) and 34.8% ($p = 0.001$) during 100–150 ms and 150–200 ms after onset of the EMG signal (Fig. 4C). No differences for $\overline{X}$RMS–EMG$_{RTD}$/M_{max} were found during any of the four time windows (Fig. 4B).

3.2. Spinal excitability

The *H* reflex could be elicited in 10 young and 11 elderly subjects. H_{max} and the H_{max}/M_{max}-ratio of the vastus medialis revealed no significant difference between groups. M_{max} was significantly reduced in the vastus lateralis (24.7%) and vastus medialis (35.6%). Elderly subjects revealed a longer Δt_M/height in the vastus medialis (7.17%) while no differences were observed in the vastus lateralis and rectus femoris. Δt_H/height re-

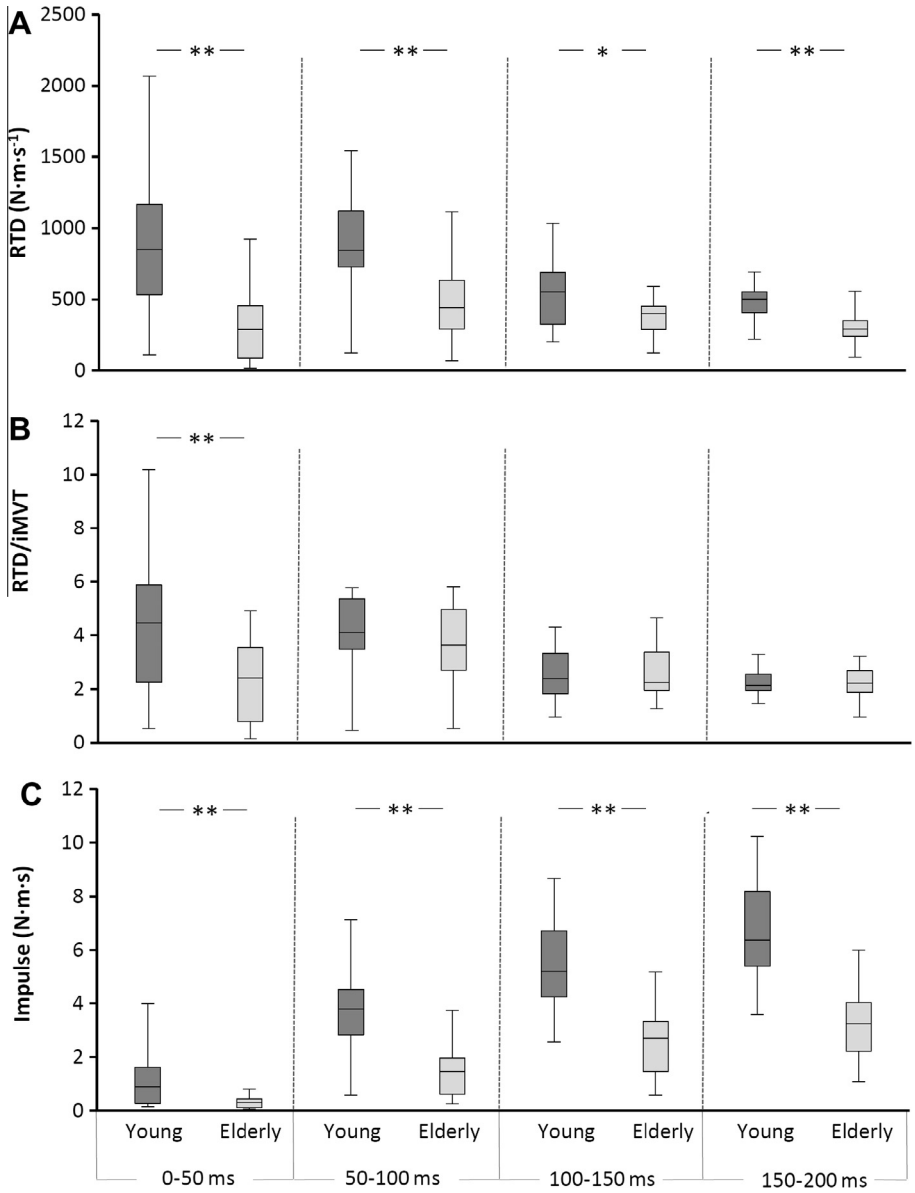

Fig. 3. Comparisons between groups: (A) rate of torque development (RTD), (B) RTD normalized to iMVT (RTD/iMVT) and (C) impulse (IMP) during the early phase of contraction in time windows of 0–50, 50–100, 100–150 and 150–200 ms after torque onset. *Denotes a significant difference between the groups (*<0.05, **<0.01).

mained unaffected by aging. The mean values ± standard deviations of evoked potentials are displayed in Table 2.

3.3. Contractile properties and lean mass

The PT, $MRTD_{TT}$ and $MRTR_{TT}$ of the resting twitch responses of the quadriceps were significantly reduced in elderly subjects (single: 61.04%, 62.9%, 64.9%, doublet: 39.9%, 53.7%, 60.3%). The TCT_{TT} of the resting twitch responses was significantly longer in elderly subjects when doublet stimulation was used (27.1%). No significant differences between the groups could be determined for $THRT_{TT}$. The TTA_{TT} of the resting twitch responses was reduced in elderly subjects (single: 50.9%, doublet: 33.6%). The mean values ± standard deviations and *p*-values of contractile properties are displayed in Table 3.

Lean body mass of the leg was significantly reduced by 19.1% in elderly subjects (Table 1).

4. Discussion

The present study was designed to provide further insight into the neuromuscular mechanisms that determine the decrease in iMVT and explosive voluntary strength of the quadriceps muscle with aging. IMVT and MRTD as well as RTD and IMP in the early phase of contraction were reduced in healthy, active elderly men and women. Voluntary activation and normalized muscle activity during iMVT were decreased indicating a reduced neural drive to the muscle. The decline in MRTD and RTD during the early phase of contraction might be more affected by modulations of muscle properties than neural changes, as corresponding muscle activity remained primarily unchanged. There were no differences in the H_{max} and H_{max}/M_{max}-ratio of the vastus medialis, but a longer normalized M_{max} latency (Δt_M/height) was observed, indicating a decline in efferent conduction velocity. In addition, the age-related decreases in iMVT and explosive voluntary strength were accompanied by changes at the muscle level, as evidenced by a decline in twitch contractile properties and leg lean mass.

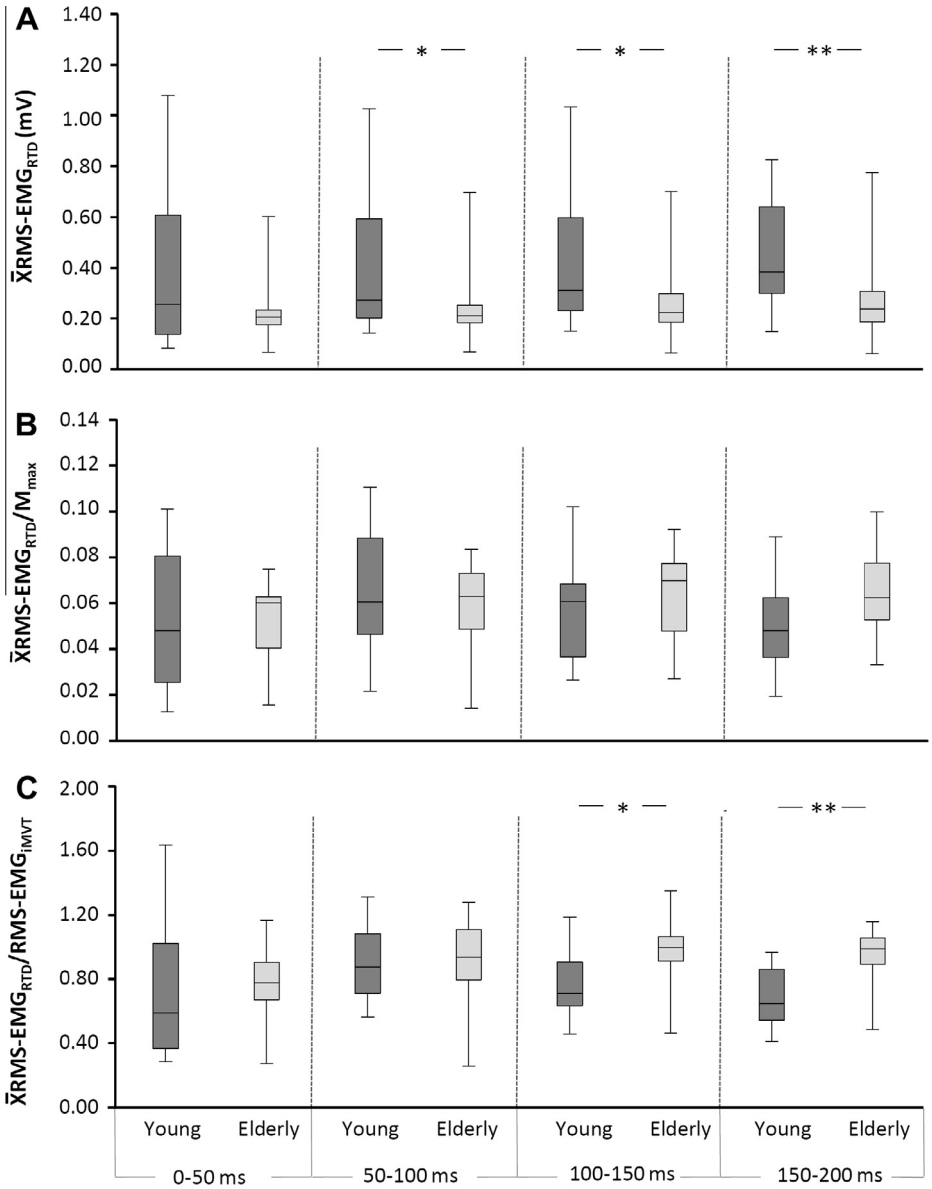

Fig. 4. Comparisons between groups of averaged root mean square of the EMG signal of vastus medialis, rectus femoris and vastus lateralis in four different time intervals after onset of the EMG signal (i.e., 0–50, 50–100, 100–150 and 150–200 ms): (A) absolute value ($\overline{X}$RMS–EMG$_{RTD}$), (B) normalized to M_{max} ($\overline{X}$RMS–EMG$_{RTD}$/M_{max}) and (C) normalized to RMS of the EMG signal during iMVT ($\overline{X}$RMS–EMG$_{RTD}$/RMS–EMG$_{iMVT}$). *Denotes a significant difference between the groups (*<0.05, **<0.01).

Table 2
Maximal M wave (M_{max}), maximal H reflex (H_{max}), H_{max}/M_{max}-ratio and height-normalized H_{max} and M_{max} latency (Δt_H/height, Δt_M/height). Values are presented as mean ± standard deviation. *Denotes a significant difference between the groups (*<0.05, **<0.01).

	Young	Elderly	p
Vastus medialis			
M_{max} (mV)	8.55 (3.04)	5.59 (3.33)	0.015*
H_{max} (mV)	1.75 (0.90)	1.57 (2.61)	0.695
H_{max}/M_{max}-ratio	0.23 (0.13)	0.34 (0.16)	0.129
Δt_H/height (ms cm^{-1})	0.105 (0.009)	0.111 (0.006)	0.065
Δt_M/height (ms cm^{-1})	0.033 (0.003)	0.036 (0.003)	0.038*
Vastus lateralis			
M_{max} (mV)	9.59 (3.27)	6.18 (3.75)	0.026*
Δt_M/height (ms cm^{-1})	0.035 (0.003)	0.037 (0.004)	0.075
Rectus femoris			
M_{max} (mV)	3.17 (0.95)	3.69 (2.01)	0.694
Δt_M/height (ms cm^{-1})	0.024 (0.003)	0.025 (0.004)	0.434

4.1. Voluntary activation

The decline in iMVT might be due to a reduced efferent motoneuron output, as confirmed by a decrease in voluntary activation and normalized muscle activity during iMVT. According to the literature, the extent to which a decrease in voluntary activation contributes to dynapenia of the quadriceps under isometric conditions remains less clear. Some authors have shown a decrease in voluntary activation (Harridge et al., 1999; Stackhouse et al., 2001; Stevens et al., 2003), whereas other investigations have observed no age-related differences (Cannon et al., 2007; Knight and Kamen, 2001; Miller et al., 2006; Roos et al., 1999; Suetta et al., 2009; Wilder and Cannon, 2009). The contrary findings might be related to divergent stimulation methods, testing procedures and the heterogeneity of the population. An age-related decrease in voluntary activation was detected when the central activation ratio was calculated (Knight and Kamen, 2001; Stackhouse et al., 2001;

Table 3
Contractile properties evaluated by single and doublet stimulation at supramaximal intensity. Peak torque (PT), maximal rate of torque development ($MRTD_{TT}$), maximal rate of torque relaxation ($MRTR_{TT}$), twitch contraction time (TCT_{TT}), twitch half relaxation time ($THRT_{TT}$) and total twitch area (TTA_{TT}) of the resting twitch torque. Values are presented as mean ± standard deviation. *Denotes a significant difference between the groups (*<0.05, **<0.01).

	Single		*p*	Doublet		*p*
	Young	Elderly		Young	Elderly	
PT (Nm)	39.39 (10.55)	15.35 (7.20)	0.000*	69.05 (14.67)	41.47 (13.91)	0.000**
$MRTD_{TT}$ (Nm s^{-1})	1814.41 (449.65)	673.54 (359.21)	0.000**	3220.57 (117.83)	1491.53 (173.61)	0.000**
$MRTR_{TT}$ (Nm s^{-1})	828.08 (171.78)	292.97 (163.94)	0.000**	1326.27 (80.82)	526.93 (42.76)	0.000**
TCT_{TT} (s)	0.064 (0.017)	0.069 (0.022)	0.101	0.067 (0.004)	0.085 (0.006)	0.013*
$THRT_{TT}$ (s)	0.084 (0.015)	0.073 (0.019)	0.106	0.090 (0.004)	0.083 (0.005)	0.290
TTA_{TT} (Nm s)	4.37 (0.55)	2.15 (0.44)	0.001**	8.70 (0.60)	5.78 (0.63)	0.006**

Stevens et al., 2003) whereas no differences could be identified when the interpolated twitch ratio was used (Cannon et al., 2007; Roos et al., 1999; Suetta et al., 2009; Wilder and Cannon, 2009). This discrepancy in results could be associated with the numbers of stimuli that were applied. The central activation ratio is classically derived by the torque generated with a pulse train. Pulse-train stimulation is more likely to cause a torque increment, and is thus regarded as more sensitive for detecting activation deficits (Miller et al., 1999). However, the present study showed an activation deficit by using paired pulses and calculating the interpolated twitch ratio whereby the standard formula (Allen et al., 1995) was corrected in order to bypass methodological limitations of the interpolated twitch technique (Folland and Williams, 2007). IMVT and the torque level just before the superimposed twitch were included in the original equation to correct the variations in the torque level (Strojnik and Komi, 1998).

The reduction of muscle activity during iMVT provided further evidence for the decreasing efferent motoneuron output to the quadriceps muscle. The only study available that has evaluated age-related changes in neural drive to the knee extensors by measuring voluntary activation and surface EMG under isometric condition was carried out by Cannon et al. (2007). These authors showed no age-related differences in either parameter between young and elderly women. However, comparing the present results with those of Cannon et al. is critical as possible gender differences might have affected the results.

In the present study, it was further shown that absolute and normalized MRTD (MRTD/iMVT) as well as absolute RTD and IMP in the early phases of torque development were reduced with aging. Absolute muscle activity was significantly decreased in elderly subjects. However, the analysis of absolute surface EMG signals is regarded as risky because various factors influence the signal and limit the comparison of data with young subjects (Klass et al., 2008). Therefore, muscle activity was normalized to M_{max} ($\overline{X}RMS–EMG_{RTD}/M_{max}$) and to muscle activity during iMVT ($\overline{X}RMS–EMG_{RTD}/RMS–EMG_{iMVT}$). Normalized EMG data suggest that the decline in explosive voluntary strength might be unrelated to changes in efferent neural drive to the quadriceps muscle. $\overline{X}RMS–EMG_{RTD}/M_{max}$ remained unchanged whereas $\overline{X}RMS–EMG_{RTD}/RMS–EMG_{iMVT}$ was significantly higher in elderly subjects in the later phases of contraction (100–150, 150–200 ms). In summary, the present data suggest that changes in skeletal muscle properties rather than neural adaptations contribute to the decrease in MRTD and RTD. This is an unexpected result, as it has been shown that aging is related to a decrease in RTD accompanied by a reduction in neural drive to the muscle (Ojanen et al., 2007). Klass et al. (2008) found that a slower RTD during the first 50 ms of torque rise was related to a reduction in motor unit discharge frequency and number of doublet discharges of the soleus muscle whereas no differences in normalized surface EMG data were observed. It was concluded that surface EMG may not always detect changes in the motor unit discharge rate. Smith et al. (1995) have shown that discharge frequency is not linearly related to surface EMG. Thus, estimating the neural drive to the muscle by surface EMG can be regarded critically (Farina et al., 2010).

Furthermore, Andersen and Aagaard (2006) observed a moderate correlation of voluntary rate of force development to twitch contractile properties. Klass et al. (2008) suggested that a greater slowing of RTD during fast voluntary contractions compared with electrically evoked twitch torques might be an indication of reduced neural activation of the muscle. The present data showed only a slight difference between MRTD and $MRTD_{TT}$ (4.9%) which indicates no age-related differences in muscular activation during RTD and thus supports the findings of normalized EMG data.

4.2. Spinal excitability

The H_{max} and the H_{max}/M_{max}-ratio of the vastus medialis remained unchanged which suggests that aging was not accompanied by differences in α-motoneuron excitability and/or presynaptic inhibition of primary muscle spindle afferents. Most studies of age-related modulations of the *H* reflex response have been carried out on the soleus muscle and have demonstrated contrary results (Kido et al., 2004; Koceja et al., 1995; Scaglioni et al., 2003). The discrepancies in results might be ascribed to differences and limitations of test conditions and stimulation protocols as well as to the variability of the reflex response which seems to enhance with increasing age (Falco et al., 1994). The measured decrease in M_{max} amplitudes is in line with findings in the literature and might be related to changes in the effectiveness of the neuromuscular junction and modifications at the muscle tissue level (Scaglioni et al., 2003). Furthermore, the analyses of normalized H_{max} latency (Δt_H/height) and Δt_M/height indicated a decrease in conduction velocity whereby only the efferent motor axons seemed to be affected. Thus, age-related changes in conduction speed, possibly due to a loss of largest axonal fibers, reduced myelination and a decreased internodal length of axons in the peripheral nerve, might contribute to the decrease in iMVT and explosive voluntary strength (Scaglioni et al., 2003).

4.3. Contractile properties and lean mass

The present results further suggest that the decrease in iMVT and explosive voluntary strength might be partially related to changes within the skeletal muscle system, since lean mass and twitch contractile properties were reduced. These results are in line with findings in the literature (Doherty, 2003; Roos et al., 1999). However, it has been demonstrated that the relationship between dynapenia and age-related loss of muscle mass (sacopenia) is weak. Thus, the contribution of other mechanisms, such as the slowing of contractile kinetics and/or neural modulations, seems to be greater in mediating age-related loss in muscle strength (Manini and Clark, 2011).

In conclusion, the data from this study confirm that the age-related decline in iMVT of the quadriceps muscle might be due to modulations of the nervous system and changes in skeletal muscle properties. Surprisingly, the reduction in MRTD as well as RTD and IMP during the early contraction phases might be more affected by modulations at the muscle level than by neural changes.

Acknowledgement

This study was supported by a scholarship of the doctorate program "Aging of individuals and society" of the Interdisciplinary Faculty, University of Rostock.

References

Allen GM, Gandevia SC, McKenzie DK. Reliability of measurements of muscle strength and voluntary activation using twitch interpolation. Muscle Nerve 1995;18(6):593–600.
Andersen LL, Aagaard P. Influence of maximal muscle strength and intrinsic muscle contractile properties on contractile rate of force development. Eur J Appl Physiol 2006;96(1):46–52.
Barbeau H, Marchand-Pauvert V, Meunier S, Nicolas G, Pierrot-Deseilligny E. Posture-related changes in heteronymous recurrent inhibition from quadriceps to ankle muscles in humans. Exp Brain Res 2000;130(3):345–61.
Behrens M, Mau-Moeller A, Bruhn S. Effect of exercise-induced muscle damage on neuromuscular function of the quadriceps muscle. Int J Sports Med 2012;33(8):600–6.
Cannon J, Kay D, Tarpenning KM, Marino FE. Comparative effects of resistance training on peak isometric torque, muscle hypertrophy, voluntary activation and surface EMG between young and elderly women. Clin Physiol Funct Imaging 2007;27(2):91–100.
Clark BC, Taylor JL. Age-related changes in motor cortical properties and voluntary activation of skeletal muscle. Curr Aging Sci 2011;4(3):192–9.
Doherty TJ. Invited review: aging and sarcopenia. J Appl Physiol 2003;95(4):1717–27.
Falco FJ, Hennessey WJ, Goldberg G, Braddom RL. H reflex latency in the healthy elderly. Muscle Nerve 1994;17(2):161–7.
Farina D, Holobar A, Merletti R, Enoka RM. Decoding the neural drive to muscles from the surface electromyogram. Clin Neurophysiol 2010;121(10):1616–23.
Folland JP, Williams AG. Methodological issues with the interpolated twitch technique. J Electromyogr Kinesiol 2007;17(3):317–27.
Folland JP, Wakamatsu T, Fimland MS. The influence of maximal isometric activity on twitch and *H*-reflex potentiation, and quadriceps femoris performance. Eur J Appl Physiol 2008;104(4):739–48.
Guiheneuc P, Bathien N. Two patterns of results in polyneuropathies investigated with the *H* reflex. correlation between proximal and distal conduction velocities. J Neurol Sci 1976;30(1):83–94.
Harridge SD, Kryger A, Stensgaard A. Knee extensor strength, activation, and size in very elderly people following strength training. Muscle Nerve 1999;22(7):831–9.
Kido A, Tanaka N, Stein RB. Spinal excitation and inhibition decrease as humans age. Can J Physiol Pharmacol 2004;82(4):238–48.
Kim J, Wang Z, Heymsfield SB, Baumgartner RN, Gallagher D. Total-body skeletal muscle mass: estimation by a new dual-energy X-ray absorptiometry method. Am J Clin Nutr 2002;76(2):378–83.
Klass M, Baudry S, Duchateau J. Voluntary activation during maximal contraction with advancing age: a brief review. Eur J Appl Physiol 2007;100(5):543–51.
Klass M, Baudry S, Duchateau J. Age-related decline in rate of torque development is accompanied by lower maximal motor unit discharge frequency during fast contractions. J Appl Physiol 2008;104(3):739–46.
Knight CA, Kamen G. Adaptations in muscular activation of the knee extensor muscles with strength training in young and older adults. J Electromyogr Kinesiol 2001;11(6):405–12.
Koceja DM, Markus CA, Trimble MH. Postural modulation of the soleus *H* reflex in young and old subjects. Electroencephalogr Clin Neurophysiol 1995;97(6):387–93.
Levine JA, Abboud L, Barry M, Reed JE, Sheedy PF, Jensen MD. Measuring leg muscle and fat mass in humans: comparison of CT and dual-energy X-ray absorptiometry. J Appl Physiol 2000;88(2):452–6.
Lohman M, Tallroth K, Kettunen JA, Marttinen MT. Reproducibility of dual-energy X-ray absorptiometry total and regional body composition measurements using different scanning positions and definitions of regions. Metabolism 2009;58(11):1663–8.
Lord SR, Ward JA, Williams P, Anstey KJ. Physiological factors associated with falls in older community-dwelling women. J Am Geriatr Soc 1994;42(10):1110–7.
Manini TM, Clark BC. Dynapenia and aging: an update. J Gerontol: Biol Sci Med Sci 2011;67(1):28–40.
Miller M, Downham D, Lexell J. Superimposed single impulse and pulse train electrical stimulation: a quantitative assessment during submaximal isometric knee extension in young, healthy men. Muscle Nerve 1999;22(8):1038–46.
Miller M, Flansbjer UB, Downham D, Lexell J. Superimposed electrical stimulation: assessment of voluntary activation and perceived discomfort in healthy, moderately active older and younger women and men. Am J Phys Med Rehabil 2006;85(12):945–50.
Murdock GH, Hubley-Kozey CL. Effect of a high intensity quadriceps fatigue protocol on knee joint mechanics and muscle activation during gait in young adults. Eur J Appl Physiol 2012;112(2):439–49.
Newman AB, Kupelian V, Visser M, Simonsick EM, Goodpaster BH, Kritchevsky SB, et al. Strength, but not muscle mass, is associated with mortality in the health, aging and body composition study cohort. J Gerontol: Biol Sci Med Sci 2006;61(1):72–7.
Ojanen T, Rauhala T, Hakkinen K. Strength and power profiles of the lower and upper extremities in master throwers at different ages. J Strength Cond Res 2007;21(1):216–22.
Porter MM, Myint A, Kramer JF, Vandervoort AA. Concentric and eccentric knee extension strength in older and younger men and women. Can J Appl Physiol 1995;20(4):429–39.
Roos MR, Rice CL, Connelly DM, Vandervoort AA. Quadriceps muscle strength, contractile properties, and motor unit firing rates in young and old men. Muscle Nerve 1999;22(8):1094–103.
Scaglioni G, Narici MV, Maffiuletti NA, Pensini M, Martin A. Effect of ageing on the electrical and mechanical properties of human soleus motor units activated by the *H* reflex and *M* wave. J Physiol 2003;548(Pt 2):649–61.
Schultz AB. Muscle function and mobility biomechanics in the elderly: an overview of some recent research. J Gerontol: Biol Sci Med Sci 1995;50 [Spec No60-3].
Smith L, Zhong T, Bawa P. Nonlinear behaviour of human motoneurons. Can J Physiol Pharmacol 1995;73(1):113–23.
Stackhouse SK, Stevens JE, Lee SC, Pearce KM, Snyder-Mackler L, Binder-Macleod SA. Maximum voluntary activation in nonfatigued and fatigued muscle of young and elderly individuals. Phys Ther 2001;81(5):1102–9.
Stevens JE, Stackhouse SK, Binder-Macleod SA, Snyder-Mackler L. Are voluntary muscle activation deficits in older adults meaningful? Muscle Nerve 2003;27(1):99–101.
Strojnik V, Komi PV. Neuromuscular fatigue after maximal stretch-shortening cycle exercise. J Appl Physiol 1998;84(1):344–50.
Suetta C, Aagaard P, Magnusson SP, Andersen LL, Sipila S, Rosted A, et al. Muscle size, neuromuscular activation, and rapid force characteristics in elderly men and women: effects of unilateral long-term disuse due to hip-osteoarthritis. J Appl Physiol 2007;102(3):942–8.
Suetta C, Hvid LG, Justesen L, Christensen U, Neergaard K, Simonsen L, et al. Effects of aging on human skeletal muscle after immobilization and retraining. J Appl Physiol 2009;107(4):1172–80.
Wilder MR, Cannon J. Effect of age on muscle activation and twitch properties during static and dynamic actions. Muscle Nerve 2009;39(5):683–91.
Zehr PE. Considerations for use of the Hoffmann reflex in exercise studies. Eur J Appl Physiol 2002;86(6):455–68.

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